A very short introduction to risk management

Project ST959 Financial Statistics

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In these slides, we briefly present the following:

- Portfolio Value, profit, loss
- Risk measures
  - Definition, Properties,...
  - Examples
    - Value at Risk
- Value at Risk (VaR) in detail
- Two approaches used by practitioners (historical and Monte Carlo)

Portfolio Value, profit, loss

Consider a portfolio.

- Let V_t denote the value (or wealth) of the portfolio at time t.

  $\longrightarrow$ We assume that V_t is a random variable on $(\Omega, \mathcal{F}, P)$.
- $V_{t+\Delta}$ is the value of the portfolio at time $t+\Delta$.
  $\longrightarrow$ $V_{t+\Delta}$ is also a random variable.
- The difference $V_{t+\Delta} - V_t$ corresponds to the profit over the time period from t to $t+\Delta$.
- We set $L_{t+\Delta} := -(V_{t+\Delta} - V_t)$.
  $\longrightarrow$ The r.v. $L_{t+\Delta}$ corresponds to the loss over the time period from t to $t+\Delta$.
- *Terminology:* The probability distribution of the r.v. $L_{t+\Delta}$ is called the loss distribution.

Please to pay attention to the sign convention when applying risk measures!

Risk measures

Idea: A risk measure is a mapping which associates to a loss L a real number (which indicates its riskiness).

Why are risk measures useful?

- Determination of risk capital: determine the amount of capital a financial institution needs to put aside to cover unexpected losses.
- Management tool: risk measures are used by management to limit the amount of risk a unit within the firm may take.
- Insurance premiums:
  compensation paid by the insured to the insurance company are based on measure of risk of the insured claims.

We will now introduce the notions of risk measure, monetary risk measure, convex risk measure and coherent risk measure.

An important theoretical contribution is that of Artzner, Delbaen, Eber, Heath (1999, journal Mathematical Finance).

We consider a one-period framework.
We thus have two dates: beginning and end of the period.

The loss at the end of the period is denoted by L.

Recall that this is a random variable on $(\Omega, \mathcal{F}, P)$.

We denote by $\mathcal{M}$ be the set of all random variables.

Note that $\mathcal{M}$ is a linear vector space.

Risk measure

Definition

A mapping $\rho : \mathcal{M} \to \mathbb{R}$ *is called risk measure if it satisfies:*

- *(A1) (Monotonicity) If* $L^1 \leq L^2$*, then* $\rho(L^1) \leq \rho(L^2)$*.*

What is the financial interpretation of the property of monotonicity?

Monetary Risk measure

Definition

A mapping $\rho : \mathcal{M} \to \mathbb{R}$ *is called a monetary risk measure if it satisfies:*

- *(A1) (Monotonicity) If* $L^1 \leq L^2$*, then* $\rho(L^1) \leq \rho(L^2)$*.*
- *(A2) (Translation invariance) For every* $L \in \mathcal{M}$*, for every* $m \in \mathbb{R}$*,* $\rho(L+m) = \rho(L) + m$*.*

Terminology: The property of translation invariance is also known as cash invariance.

What is the financial interpretation of the property of translation invariance?

Convex risk measure

Definition

A mapping $\rho : \mathcal{M} \to \mathbb{R}$ *is called convex risk measure if it satisfies :*

- *(A1) (Monotonicity) If* $L^1 \leq L^2$*, then* $\rho(L^1) \leq \rho(L^2)$.
- *(A2) (Translation invariance) For every* $L \in \mathcal{M}$*, for every* $m \in \mathbb{R}$*,* $\rho(L+m) = \rho(L) + m$.
- *(A3) (Convexity) For every* $L^1, L^2 \in \mathcal{M}$ *and every* $\lambda \in [0,1]$*,*

$$\rho(\lambda L^1 + (1-\lambda)L^2) \leq \lambda\rho(L^1) + (1-\lambda)\rho(L^2).$$

What is the financial interpretation of the property of convexity?

Coherent risk measure

Definition

A mapping $\rho : \mathcal{M} \to \mathbb{R}$ *is called coherent risk measure if it satisfies:*

- *(A1) (Monotonicity) If* $L^1 \leq L^2$*, then* $\rho(L^1) \leq \rho(L^2)$.
- *(A2) (Translation invariance) For every* $L \in \mathcal{M}$*,* $m \in \mathbb{R}$*,* $\rho(L+m) = \rho(L) + m$.
- *(A3) (Convexity) For every* $L^1, L^2 \in \mathcal{M}$ *and every* $\lambda \in [0,1]$*,*

$$\rho(\lambda L^1 + (1-\lambda)L^2) \leq \lambda\rho(L^1) + (1-\lambda)\rho(L^2).$$

- *(A4) (Positive homogeneity) For every* $L \in \mathcal{M}$*, for every* ***positive*** *number* c*,*

$$\rho(cL) = c\rho(L).$$

What is the financial interpretation of positive homogeneity?

What is a law-invariant risk measure?
(Also, known as distribution-invariant risk measure).

First (very easy) example of a risk measure

Let us consider $\rho : \mathcal{M} \to \mathbb{R}$ defined by

$$\rho(L) = E(L).$$

In order for ρ to be well-defined in this example,
we restrict ourselves to random variables L having a finite expectation.

Questions:
Is the mapping ρ in this example a risk measure?
Is ρ a monetary risk measure?
Is ρ convex?
Is ρ coherent?

Value at Risk (VaR)

Let us fix a given level $\alpha \in (0,1)$ (known as the confidence level).

In practice, $\alpha = 95\%$, $\alpha = 99\%$, $\alpha = 99.9\%$.

The Value at Risk at confidence level α (denoted by VaR_α) is the smallest number l such that the probability that the loss L exceeds l is no larger than $(1-\alpha)$.

More precisely, we have the following definition:

$$\mathrm{VaR}_\alpha(L) = \inf\{l \in \mathbb{R} : P(L > l) \leq 1 - \alpha\}.$$

We will now provide a link between **VaR** of L and **the quantile function** of L.

For this, let us recall the definition of the quantile function and some properties.

Quantile function (Revision)

Let X be a random variable.
Let F_X be a the cumulative distribution function of the r.v. X.

Definition

The quantile function q_X of X is defined by:

$$q_X(y) = \inf\{x \in \mathbb{R} : F_X(x) \geq y\}, \text{ for all } y \in (0,1).$$

Terminology: The function q_X is also known as *the lower quantile function* of X, or the left-continuous quantile function of X.

Terminology: Fix a point $y \in (0,1)$. The number $q_X(y)$ is known as the quantile of X at level y, or the y-quantile of X.

Quantile function (Revision)

Some properties:

- The quantile function q_X is non-decreasing.
- The quantile function q_X is left-continuous with right limits.

Quantile function (Revision)

Proposition
(The case where F is increasing and continuous)

Let F_X be a given CDF.
We assume that F_X is **increasing and continuous**.
Then,

- F_X is an **invertible** function.
- Moreover,
$$q_X(y) = F_X^{-1}(y), \text{ for all } y \in (0,1),$$
where F_X^{-1} denotes the (usual) inverse function of F_X.

Back to the VaR

Recall the definition of the VaR at level α:

$$\mathrm{VaR}_\alpha(L) = \inf\{l \in \mathbb{R} : P(L > l) \leq 1 - \alpha\}.$$

- Let us denote by F_L the CDF of L,
  that is, $F_L(x) = P(L \leq x)$, for all $x \in \mathbb{R}$.
  Then we have

$$\begin{aligned}\mathrm{VaR}_\alpha(L) &= \inf\{l \in \mathbb{R} : P(L > l) \leq 1 - \alpha\} \\ &= \inf\{l \in \mathbb{R} : F_L(l) \geq \alpha\} \\ &= q_L(\alpha).\end{aligned}$$

  Conclusion: The VaR of the loss L at level α is equal to the α-quantile of the loss L.
  $\longrightarrow$ Practical advice: It is useful to know properties of quantile functions, as we can easily deduce properties of the VaR.

Question: Is VaR_α a risk measure?

Theorem

Let $\alpha \in (0,1)$ be a fixed confidence level.
We have:

- (Monotonicity) If $L^1 \leq L^2$, then $VaR_\alpha(L^1) \leq VaR_\alpha(L^2)$.
- (Translation invariance) For every r.v. L, for every $m \in \mathbb{R}$, $VaR_\alpha(L+m) = VaR_\alpha(L) + m$.
- (Positive Homogeneity) For every r.v. L, for every **positive** number c,
$$VaR_\alpha(cL) = cVaR_\alpha(L).$$

Caveat: In general, VaR_α is **not convex!**

.

Remember:

VaR_α is a monetary risk measure which satisfies positive homogeneity.

But, VaR_α is **not a convex risk measure** (as it does not satisfy the property of convexity).

- Advantage: VaR_α is simple to use
  (it is just the $\alpha -$ *quantile* of the distribution of the loss).
- Drawback: VaR does not provide information about the size of the losses which might occur with probability less than $1 - \alpha$.
- Drawback: Negative diversification effects can arise
  (as VaR_α is not convex).

VaR_α is widely used by practitioners.

What does the EU insurance regulatory framework Solvency II say about Value at Risk?

Particular case: VaR in the case of normal distribution

Let $m \in \mathbb{R}$ and $\sigma > 0$.

Property

If $L \sim N(m, \sigma^2)$, then

$$VaR_\alpha(L) = m + \sigma\Phi^{-1}(\alpha).$$

Here (as usual), Φ denotes the CDF of the standard normal $N(0,1)$.

Quantile function (Revision)

Proposition
(Quantile transformation of a uniform random variable)

Let U be a r.v. such that $U \sim U(0,1)$ (uniform distribution on the interval $(0,1)$).
Let X be a random variable with quantile function q_X.
Set $Y = q_X(U)$.
Then, the random variable Y and the random variable X have the same distribution. That is,

$$F_Y(x) = F_{q_X(U)}(x) = F_X(x), \text{ for all } x \in \mathbb{R}.$$

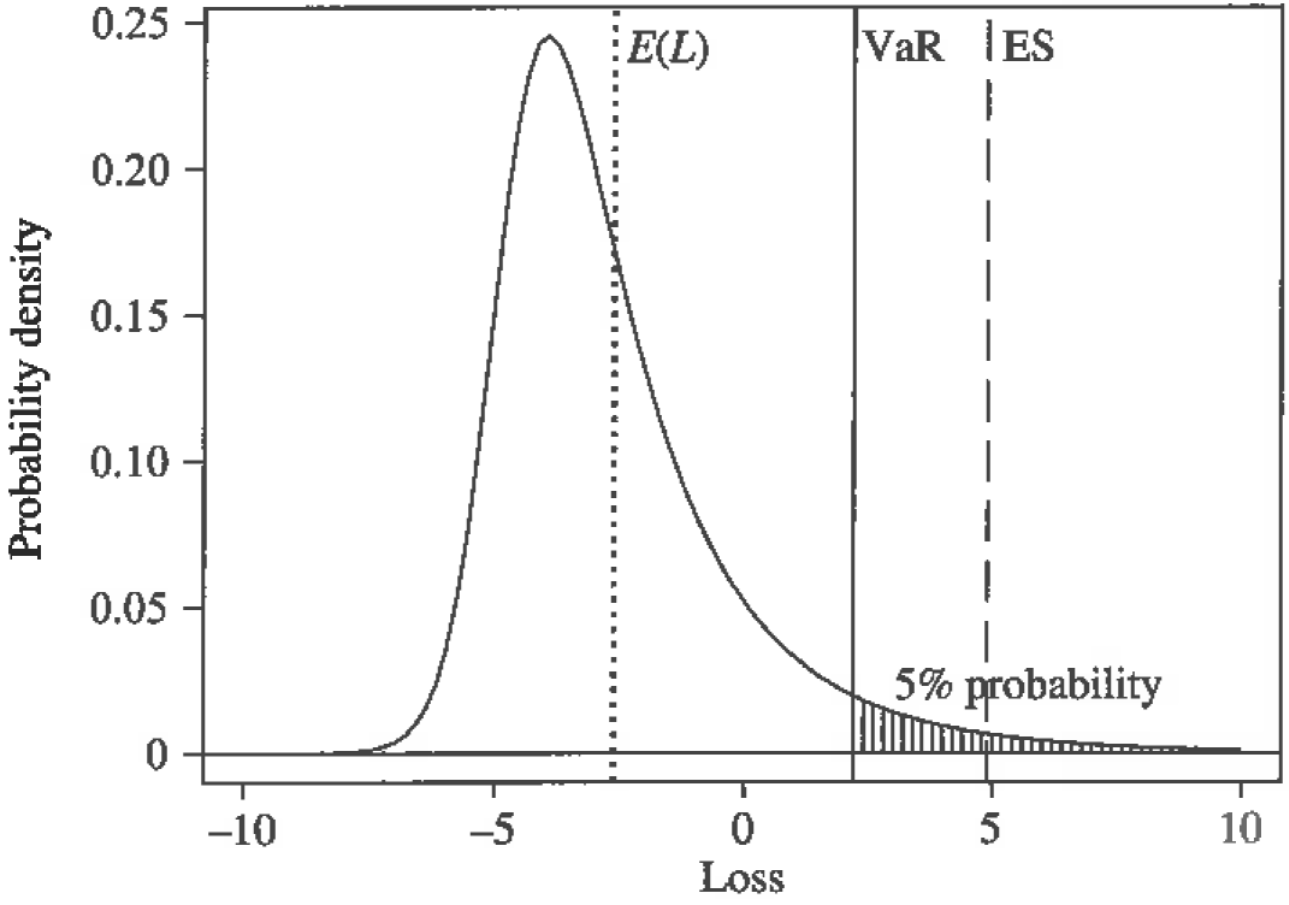
This proposition is very useful for **simulating** from a given distribution.

Quantile function (Revision)

Proposition

Let X be a random variable.
Let F_X be a the cumulative distribution function of the r.v. X.
Assume moreover that the CDF F_X is a **continuous** function.
Set $U = F_X(X)$. Then, the random variable $U = F_X(X)$ has the Uniform distribution $U(0, 1)$.

An example through a picture



Figure

ES stands for Expected Shortfall (we have not presented it here).

Approaches for predicting VaR over the next period

Here, we will give some "recipes" used by practitioners.

- Historical estimation
- Monte Carlo simulation

It is important that we understand the underlying assumptions, the main steps, the advantages and disadvantages of each method.

We follow the presentation from the book by Embrechts, Frey, McNeil and more information can be found in this reference.

We will present the situation when we have only one factor which affects the value of the portfolio (and, hence, the loss).

This is typically the case when we trade with one stock only (as opposed to a portfolio with several different stocks).

Historical Estimation

Idea: This method uses historical data for the risk factor changes and an estimate for the probability distribution of the loss based on the empirical distribution.

There are no assumptions on the probability distribution of X made: this is a **non-parametric** method.

Historical Estimation

We place ourselves at t.

The main steps of this method are:

- Collect a historical dataset $X_{t-(k-1)\Delta}, \ldots, X_{t-2\Delta}, X_{t-\Delta}, X_t$.

  Remark: Note that k corresponds to the size of the data set.

- Use the above data to "get" a dataset of losses:

$$\widetilde{L}_t = l_{[t]}(X_t), \widetilde{L}_{t-\Delta} = l_{[t]}(X_{t-\Delta}), \ldots, \widetilde{L}_{t-(k-1)\Delta} = l_{[t]}(X_{t-(k-1)\Delta}),$$

  where $l_{[t]}$ is the loss operator at time t.

  Question: What is the expression of the loss operator in our case?

Remark: $\widetilde{L}_{t-\Delta}$ corresponds to the portfolio loss that we would have if the risk factor changes observed between $t-2\Delta$ and $t-\Delta$ were to recur.

- Use this dataset to compute an empirical distribution estimate (e.g. "histogram estimate") of the loss probability distribution.

Historical Estimation

- Compute an estimate for $VaR_\alpha(L_{t+\Delta})$.

  An estimate for $VaR_\alpha(L_{t+\Delta})$ can be obtained as follows:
  - Order the data points of the set from the smallest loss to the greatest loss:
    $\widetilde{L}^{(1)} \leq \widetilde{L}^{(2)} \leq \ldots \leq \widetilde{L}^{(k)}$.
  - An estimate for $VaR_\alpha(L_{t+\Delta})$ is given by:

$$\widetilde{L}^{([k\alpha])},$$

  where $[k\alpha]$ denotes the largest integer not exceeding $k\alpha$.

Historical Estimation

- Advantages:
  Easy to implement.
  No assumption about distribution and dependence structure of risk-factors.
- Disadvantages:
  -In this approach, "the worst case is never worse than what has happened in the past".
  -Need of a large sample of relevant historical data to get reliable estimates of VaR_α; missing data can cause problems.

Monte Carlo simulation

Monte Carlo simulation (in risk management) is a general name for any approach which involves simulation from an explicit parametric model for the risk factor changes.

Main steps of this approach:

- Choose a parametric model for the risk factor changes X.
- Use historical data $X_{t-(k-1)\Delta}, \ldots, X_{t-2\Delta}, X_{t-\Delta}, X_t$ of past risk factor changes to estimate the parameters of the model.
- Use the model from the previous point to simulate M (possible) values of risk factor changes for the next period (that is, the period from t to $t+\Delta$):
$$\widetilde{X}^1_{t+\Delta}, \widetilde{X}^2_{t+\Delta}, \ldots, \widetilde{X}^M_{t+\Delta}.$$
- Use the simulated risk factors to simulate losses from the period from t to $t+\Delta$ using the loss operator $l_{[t]}$.
- From here, compute VaR_α as in the previous method.

Monte Carlo simulation

- Advantages: We can choose the number of simulations M ourselves
  (M can be chosen much bigger than k).
- Disadvantages: simulation can be computationally intensive; "results are as good as is the model used" (think of model risk).